



Kharcha

WALLET

Smart Expense Management

KHARCHA WALLET

Smart Expense Management System

◆ PHP + MySQL Web Application ◆

Project Type	Website — Expense Management System
Project Name	Kharcha Wallet
Technology	PHP (Hypertext Preprocessor)
Database	MySQL
Submitted By	Jariya Vishvjeet
Academic Year	2026-'27

◆ 1. PROJECT ABSTRACT

Kharcha Wallet is a feature-rich, PHP-based web application designed to empower individuals and families to efficiently track, manage, and analyze their daily expenses. The term "Kharcha" (खर्चा) means expenditure in Hindi, making this platform culturally relevant and intuitive for a wide user base.

The system provides a comprehensive dashboard that enables users to record income and expenses, categorize transactions, set monthly budgets, generate visual reports, and receive automated budget alerts — all through a clean, responsive web interface accessible from any device.

Key Highlights

- Real-time expense tracking with category-wise breakdown
- Interactive charts for monthly and annual spending analysis
- Budget management with automated overspending notifications
- Secure user authentication with role-based access (User/Admin)
- PDF/CSV report export for financial record keeping
- Responsive design — accessible on desktop, tablet, and mobile
- Built on PHP 8.x with MySQL database for robust data management



Project Objective

To develop a secure, scalable, and user-friendly web-based expense management system that helps users gain visibility over their finances, encourage disciplined spending habits, and make informed financial decisions through data-driven insights.

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2. WEBSITE DESCRIPTION

2.1 Overview

Kharcha Wallet is a multi-module web application developed using PHP (server-side scripting) and MySQL (relational database). The front-end uses HTML5, CSS3, and JavaScript with Bootstrap for responsive layouts. The back-end handles all business logic including authentication, CRUD operations on expenses, budget calculation, and report generation.

2.2 Core Modules

Module	Description
 Authentication	Secure login/logout, registration with hashed passwords (bcrypt), session management
 Dashboard	Summary cards showing total income, expenses, savings, and real-time graphs
 Expense Manager	Add, edit, delete expenses with date, category, amount, and notes
 Categories	Custom expense categories with icons and color labels
 Reports	Monthly/yearly reports with Chart.js pie and bar graphs, export to PDF/CSV
 Alerts	Automated email/in-app notifications when budget limits are exceeded
 Admin Panel	User management, system settings, data backup and restore

2.3 Technology Stack

- PHP 8.x — Server-side backend logic and routing
- MySQL 8.x — Relational database for all data storage
- HTML5 + CSS3 — Front-end structure and styling
- Bootstrap 5 — Responsive layout and UI components
- JavaScript / jQuery — Dynamic interactions and AJAX calls
- Chart.js — Interactive financial charts and graphs
- PHPMailer — Email notification services

3. FOLDER STRUCTURE

The project follows an MVC-inspired directory layout for clean separation of concerns:

Path / File	Purpose
kharcha-wallet/	Root project directory
— index.php	Application entry point & router
— config/	Configuration files directory
— db.php	Database connection (PDO/MySQLi)
— config.php	App settings, constants, keys
— controllers/	Business logic controllers
— AuthController.php	Login, Register, Logout
— ExpenseController.php	CRUD for expenses
— ReportController.php	Report generation
— AdminController.php	Admin operations
— models/	Database model classes
— User.php	User data model
— Expense.php	Expense data model
— Category.php	Category data model
— Budget.php	Budget data model
— views/	HTML/PHP view templates
— auth/	Login & Register pages
— dashboard/	Main dashboard view
— expenses/	Expense list & forms
— reports/	Report views & charts
— admin/	Admin panel views
— assets/	Static front-end assets
— css/	Custom stylesheets
— js/	JavaScript files
— images/	Icons and images
— includes/	Shared PHP components
— header.php	Common header/navbar

─ footer.php	Common footer
─ functions.php	Helper functions
─ sql/	Database files
─ kharcha_wallet.sql	Full DB schema & seed data
─ README.md	Project documentation

4. DATA FLOW DIAGRAM (DFD)

A Data Flow Diagram (DFD) illustrates how data flows through the Kharcha Wallet system, from user inputs through processing to data stores and outputs.

4.1 DFD Level 0 — Context Diagram

The context diagram shows the system as a single process interacting with external entities: Users, Database, Payment systems, and Report modules.

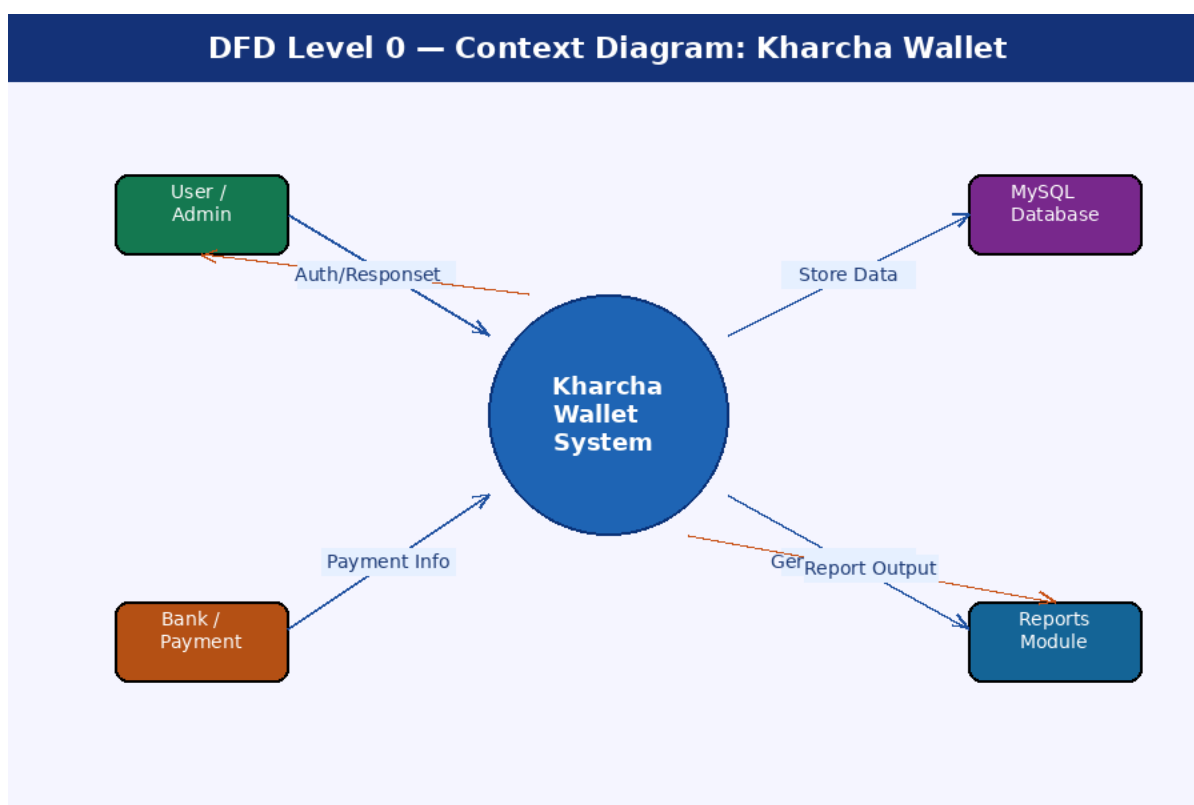


Figure 4.1 — DFD Level 0: Context Diagram — Kharcha Wallet System

4.2 DFD Level 1 — Detailed Processes

Level 1 expands the central process into 6 sub-processes: User Authentication (P1), Add Expense (P2), View Reports (P3), Manage Categories (P4), Budget Alerts (P5), and Admin Panel (P6).

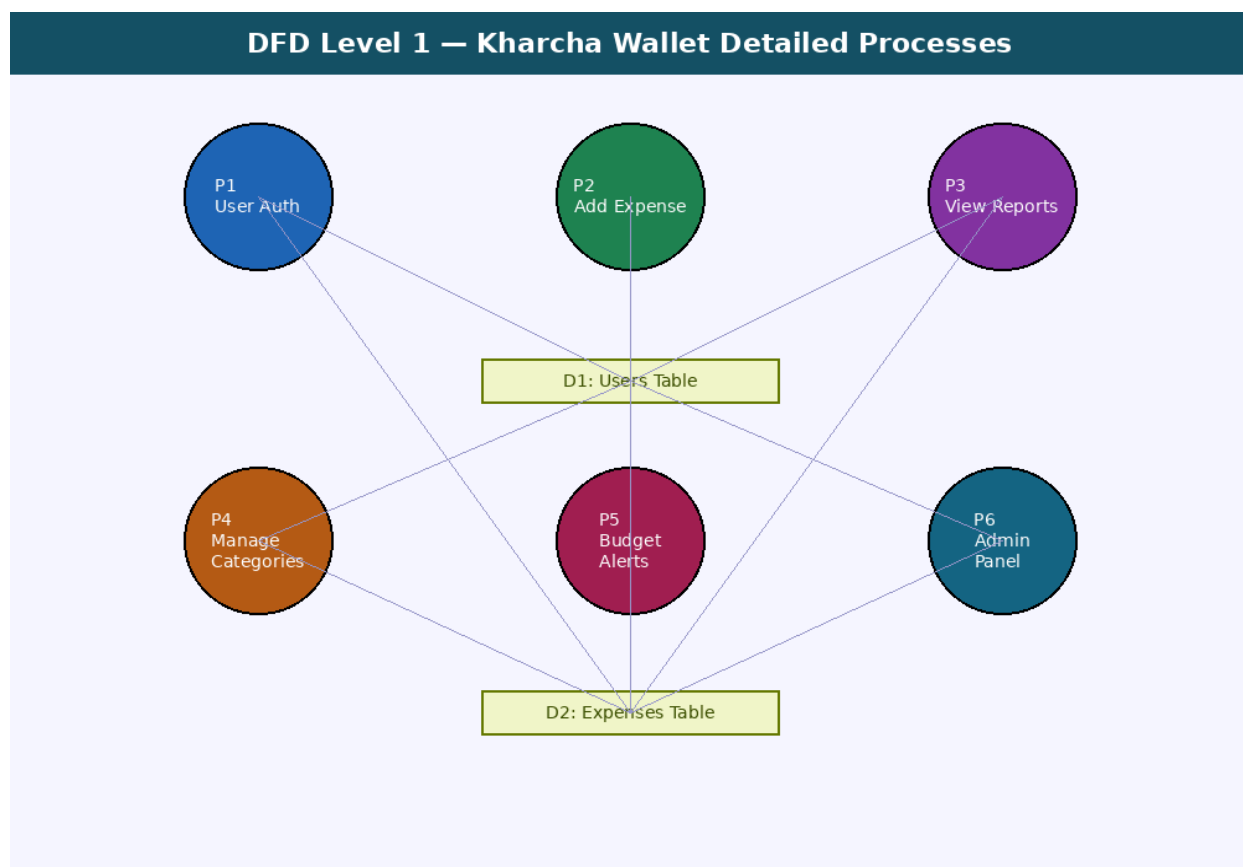


Figure 4.2 — DFD Level 1: Detailed Process Decomposition

5. ENTITY-RELATIONSHIP (ER) DIAGRAM

The ER Diagram represents the logical structure of the Kharcha Wallet MySQL database, showing all entities, their attributes, primary keys (PK), foreign keys (FK), and relationships between tables.

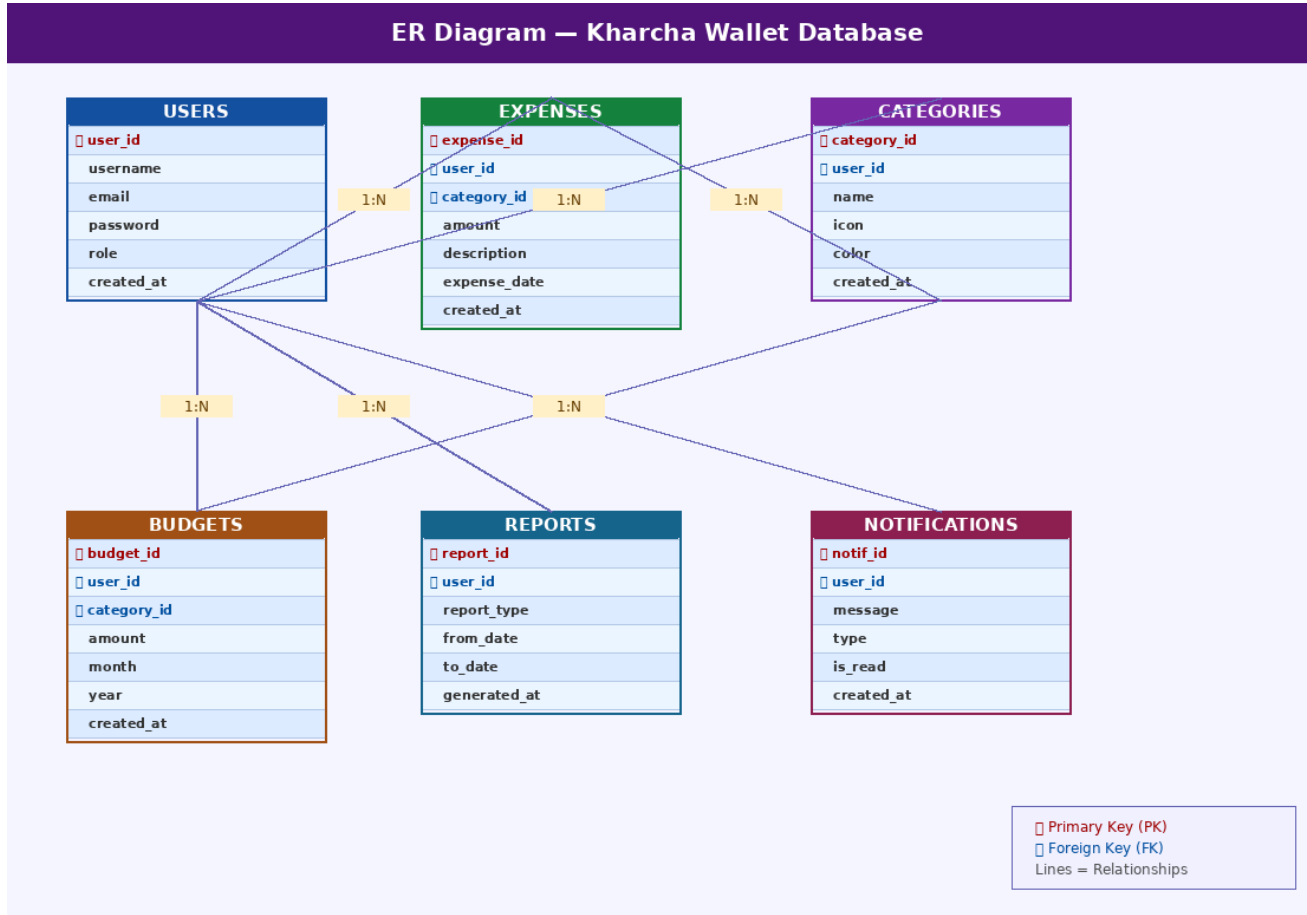


Figure 5.1 — ER Diagram: Kharcha Wallet Database Schema (PK = Primary Key, FK = Foreign Key)

5.1 Relationships Summary

- USERS → EXPENSES : One user can have many expenses (1:N)
- USERS → CATEGORIES : One user can create many categories (1:N)
- CATEGORIES → EXPENSES : One category can contain many expenses (1:N)
- USERS → BUDGETS : One user can set multiple budgets (1:N)
- CATEGORIES → BUDGETS : One category can have multiple budget entries (1:N)
- USERS → REPORTS : One user can generate many reports (1:N)
- USERS → NOTIFICATIONS : One user can receive many notifications (1:N)

6. DATA DICTIONARY

The Data Dictionary provides a formal description of all database tables, their fields, data types, constraints, and purposes in the Kharcha Wallet system.

6.1 Table: USERS

Field Name	Data Type	Size	Key	Description
user_id	INT	11	PK	Auto-incremented unique user identifier
username	VARCHAR	100		Display name chosen by the user
email	VARCHAR	150		Unique email address for login
password	VARCHAR	255		bcrypt-hashed password string
role	ENUM	-		User role: "user" or "admin"
created_at	DATETIME	-		Account creation timestamp

6.2 Table: EXPENSES

Field Name	Data Type	Size	Key	Description
expense_id	INT	11	PK	Unique identifier for each expense record
user_id	INT	11	FK	References USERS(user_id) — expense owner
category_id	INT	11	FK	References CATEGORIES(category_id)
amount	DECIMAL	10,2		Monetary value of the expense (INR)
description	VARCHAR	255		Brief note about the expense
expense_date	DATE	-		Date when the expense was incurred
created_at	DATETIME	-		Record creation timestamp

6.3 Table: CATEGORIES

Field Name	Data Type	Size	Key	Description
category_id	INT	11	PK	Unique category identifier
user_id	INT	11	FK	References USERS(user_id) — category owner
name	VARCHAR	100		Category label (e.g., Food, Transport)

icon	VARCHAR	50		Icon class or emoji for the category
color	VARCHAR	10		Hex color code for UI display
created_at	DATETIME	-		Category creation timestamp

6.4 Table: BUDGETS

Field Name	Data Type	Size	Key	Description
budget_id	INT	11	PK	Unique budget record identifier
user_id	INT	11	FK	References USERS(user_id)
category_id	INT	11	FK	References CATEGORIES(category_id)
amount	DECIMAL	10,2		Monthly budget limit amount (INR)
month	TINYINT	2		Budget month (1-12)
year	YEAR	4		Budget year (e.g., 2026)
created_at	DATETIME	-		Budget entry creation timestamp

6.5 Table: REPORTS

Field Name	Data Type	Size	Key	Description
report_id	INT	11	PK	Unique report identifier
user_id	INT	11	FK	References USERS(user_id)
report_type	VARCHAR	50		Type: monthly, yearly, category-wise
from_date	DATE	-		Report start date
to_date	DATE	-		Report end date
generated_at	DATETIME	-		Timestamp when report was generated

6.6 Table: NOTIFICATIONS

Field Name	Data Type	Size	Key	Description
notif_id	INT	11	PK	Unique notification record identifier
user_id	INT	11	FK	References USERS(user_id)
message	TEXT	-		Notification message content
type	VARCHAR	50		Type: budget_alert, info, warning
is_read	TINYINT	1		Read status: 0 = unread, 1 = read

created_at	DATETIME	-		Notification creation timestamp
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